

Week 5 — Confidence intervals

Business Forecasting

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Where we are

Describing data → **Reasoning under uncertainty** → Regression → Time series forecasting

Reasoning under uncertainty. A sample mean is a single number. This week we put a **range of plausible values** around it — a confidence interval.

- **This week (W5): confidence intervals** — for a **mean** (normal & Student's t) and for a **variance** (chi-square).

Fast & hands-on: every interval we derive, you build in R with `qnorm` / `qt` / `qchisq` — the exercises run **right here in the deck**, in your browser.



Before we start — two minutes on last week

`Sample_list` holds the 200 sampled Mexico City listings: `price` in pesos and `review_scores_cleanliness`, the guest cleanliness rating out of 5.

1. Compute the standard deviation of `Sample_list$review_scores_cleanliness`.
2. The sample has **200** listings. Turn that standard deviation into the standard error of the mean rating.
3. Recompute that standard error as if the sample had **800** listings instead, and report how many times smaller it is.

Exercise

Start Over

Show Hint

Run Code

```
1 # task 1: spread of the individual ratings
2
3 # task 2: standard error of the mean rating, n = 200
4
5 # task 3: the same standard error at n = 800, and the ratio
```

Confidence Intervals

Confidence intervals

- We calculated the mean price in our sample
- How confident are we that our estimate is close to the true parameter?
- A **confidence interval** measures the uncertainty around the estimate — a range of plausible values for the parameter.

Confidence intervals — a first look

- Mean price in our sample was **1245.43**
- Is it reasonable to think the true average price is 1100? What about 2000?
- Suppose we computed the confidence interval to be:

$$\{1086.64, 1404.22\}$$

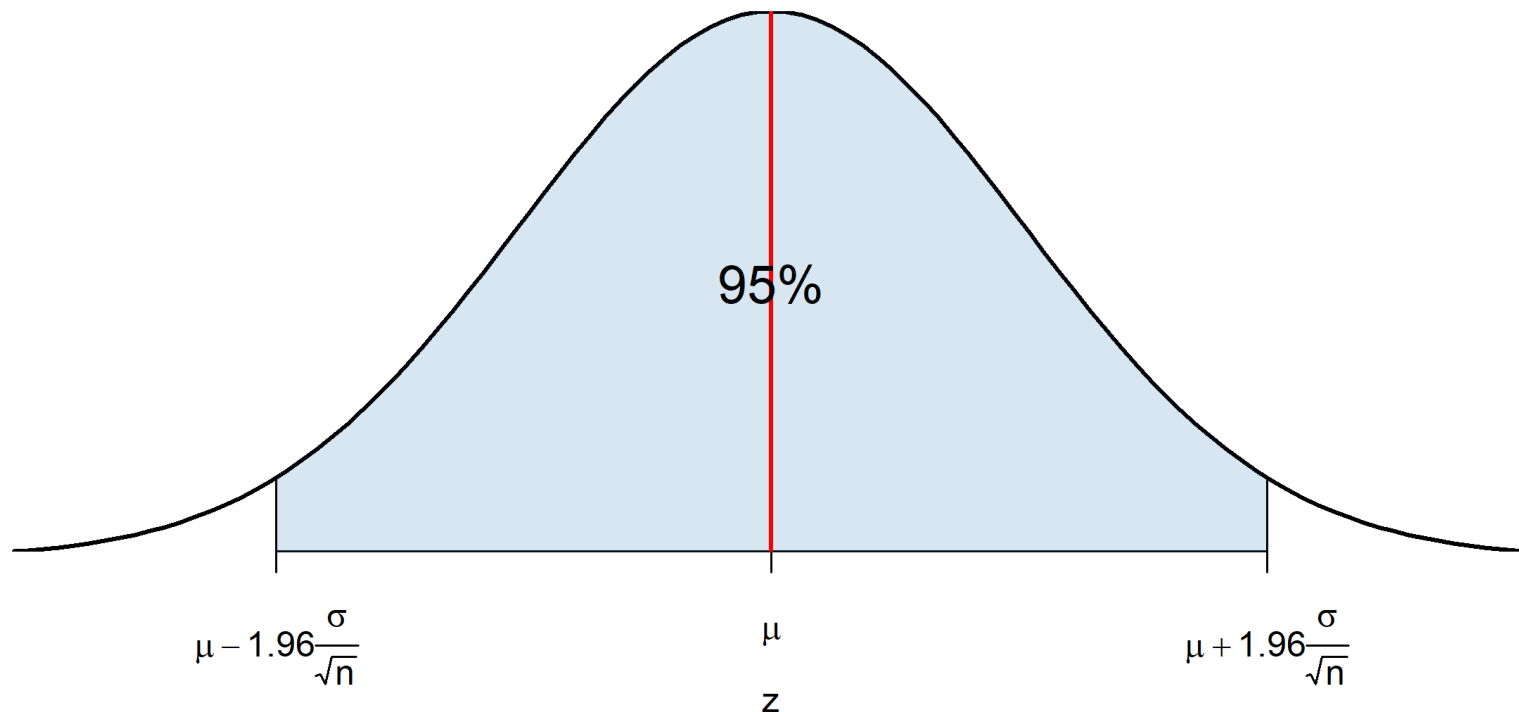
- Where do these numbers come from?
 1. The **sampling distribution** of the sample mean tells us how far a point estimate typically lands from the true mean.
 2. The confidence interval uses that to say where the true mean plausibly lies.

Sampling distribution

Q: How likely is it that a sample mean is far from the true mean?

- Recall the sampling distribution: $\bar{x} \sim \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$
- Across repeated samples, **95%** of the means land in the shaded region. Why 1.96?

Sampling Distribution of Sample Mean



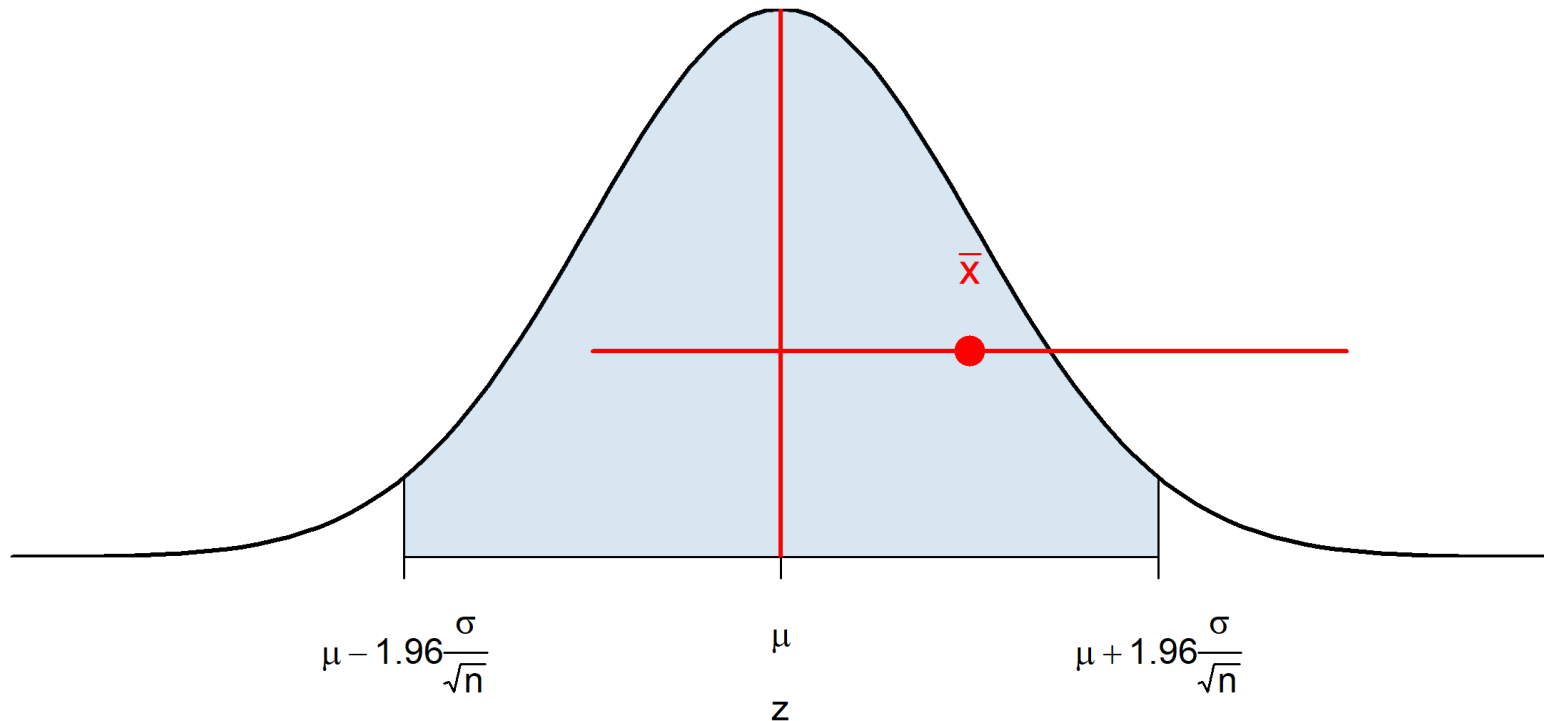
Where 1.96 comes from

$$\begin{aligned} 0.95 &= P(-1.96 < Z < 1.96) = P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{1-\frac{\alpha}{2}}\right) \\ &= P\left(\mu + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} < \bar{X} < \mu + z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right) \end{aligned}$$

- The CLT guarantees $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ is standard normal.
- What if you don't know σ ? In large samples $s \rightarrow \sigma$, so $\frac{\bar{X} - \mu}{s/\sqrt{n}} \rightarrow N(0, 1)$.
 - You may need a somewhat larger n so that s approximates σ well.

Margin of error

- So 95% of sample means fall within $1.96 \frac{\sigma}{\sqrt{n}}$ of the true mean.
- This half-width is the **margin of error**: $ME = z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}$.
- The interval $\bar{x} \pm ME$ covers μ whenever $|\mu - \bar{x}| < ME$.





Exercise 1 — Point estimate and spread

The dataset `Sample_list` (a sample of Airbnb listings) is loaded, and the nightly prices are in the vector `x`. Before any interval: get the three ingredients the formula needs — the point estimate, the spread, and the sample size.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Three numbers. Store the sample size in n.
```



Exercise 2 — How far does the *mean* move?

`sd(x)` says how much individual prices vary. That is **not** how much the sample *mean* varies. Compute the standard deviation of the sample mean — the **standard error**, $SE = s/\sqrt{n}$ — store it in `se`, and compare it to `sd(x)`.

Exercise

↻ Start Over

💡 Show Hint

▶ Run Code

```
1 # se <- ...
2 # then compare se with sd(x): how many times smaller is it?
```

Buying precision

Your interval is too wide for the pricing meeting. You can pay a survey firm to collect **four times** as many listings.

How much narrower does the interval get for that money? Write your guess down.

Exercise 3 — Margin of error, and what 4× the data buys

Compute the margin of error $ME = z_{0.975} SE$ for the sample as it is. Then compute the *hypothetical* ME you would get with **4× the sample size** and the same `sd`. Report the ratio.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 n <- length(x)
2 # me      <- ...      the margin of error now
3 # me_big  <- ...      the margin of error with 4 * n observations, same sd
4 # me / me_big          the factor by which it shrank
```

What 4× the data bought you

The margin of error scales like $1/\sqrt{n}$:

$$\text{ME} = z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}$$

- **4× the data halves the margin of error.** Not a quarter of it — half.
- To halve it again you need **16×** the original sample.
- Precision gets expensive fast: this is why survey budgets grow quadratically in the precision you promise.

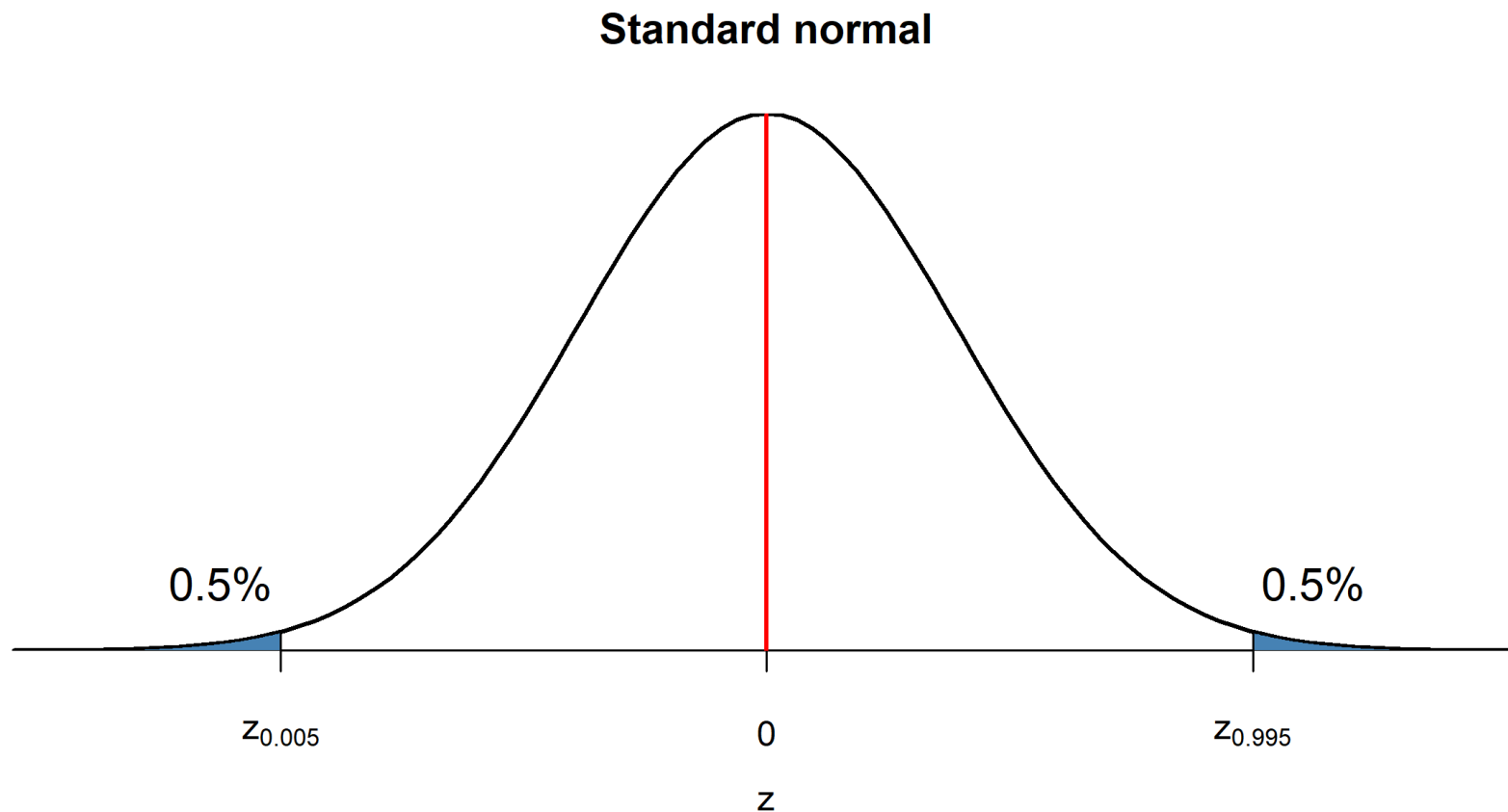
Calculation procedure

Use this when $n > 40$:

1. Take an IID sample.
2. Compute the mean $\bar{x}$ and standard deviation s .
 - **Standard error** = SD of the estimator: $SE = \frac{s}{\sqrt{n}}$.
3. Pick a **confidence level** $1 - \alpha$ (usually 90 / 95 / 99%).
 - α = probability of a Type 1 error. **Example:** 95% $\rightarrow \alpha = 0.05$.
4. Find critical values $z_{\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}}$ with $P(z_{\frac{\alpha}{2}} < Z < z_{1-\frac{\alpha}{2}}) = 1 - \alpha$.
 - **Example:** 95% $\rightarrow z_{0.975} = 1.96$.
5. Build the interval: $\left\{ \bar{x} - z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}, \bar{x} + z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}} \right\}$

Finding critical values

- Suppose the confidence level is 99%, so $\alpha = 0.01$.
- We need $z_{1-\frac{\alpha}{2}}$ with $P(z_{\frac{\alpha}{2}} < Z < z_{1-\frac{\alpha}{2}}) = 0.99$.
- Equivalently $z_{0.995}$ is the 99.5% quantile of the standard normal $\rightarrow z_{0.995} = 2.58$.





Exercise 4 — Build the 95% interval

You now have all five steps. Build the 95% confidence interval for the mean price yourself, from `x`, and print both endpoints.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Steps 2, 4, 5 of the procedure. Two endpoints, one line if you want.
```

Constructing a CI: worked example

90% CI for the average price of listings with cleanliness score > 4.5:

1. IID sample, $n = 100$ ✓
2. $\bar{x} = 1245.43$ and $s = 961.9$
3. Confidence level 90% $\rightarrow \alpha = 0.1$
4. Critical value: $z_{0.95} = 1.65$
5. Interval:

$$\left\{ \underbrace{1245.43}_{\text{sample mean}}, \underbrace{-1.65 \cdot \frac{961.9}{\sqrt{100}}}_{\text{margin of error}}, \underbrace{1245.43 + 1.65 \cdot \frac{961.9}{\sqrt{100}}}_{\text{sample mean} + \text{margin of error}} \right\} = \{1086.64, 1404.22\}$$

Predict before you compute

You are about to rebuild the same interval at **90%**, **95%** and **99%** confidence.

Guess first: how much wider is the 99% interval than the 95% one?

- Roughly the same? 30% wider? Twice as wide?

Commit to a number, then compute it.



Exercise 5 — Change the confidence level

Rebuild the interval for the mean price at **90%**, **95%** and **99%**. Report the three **widths**, and the ratio of the 99% width to the 95% width.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 n <- length(x)
2 se <- sd(x) / sqrt(n)
3
4 # One interval per confidence level. Which quantile goes with which level?
5 # ci90 <- ...
6 # ci95 <- ...
7 # ci99 <- ...
8 # then diff() each one, and take the ratio.
```

The price of confidence

Going 95% → 99% costs about **31% extra width** — the critical value goes from 1.96 to 2.58 and everything else stays fixed.

- Higher confidence buys a *safer* statement about a *vaguer* range.
- A 100% interval is $(-\infty, \infty)$: always right, never useful.
- Which level you choose is a business decision about the cost of being wrong, not a statistical one.



Exercise 6 — What does the width actually depend on?

The formula has three inputs you could change: the critical value, n , and s . Hold the confidence level at 95% and find out how the width responds to the other two.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 ns <- c(10, 25, 50, 100, 400, 1000)
2
3 # 1. Width of the 95% interval at each of those n, keeping sd(x) fixed.
4
5 # 2. Plot width against n. What shape is it?
6
7 # 3. Now keep n = 200 and double the spread: replace sd(x) with 2 * sd(x).
8 #    What happens to the width?
```

Shape of confidence intervals

$CI = \left\{ \bar{x} \pm z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}} \right\}$ is **wider** when:

- the **confidence level** is higher (99% is wider than 90%);
- the sample size n is **smaller**;
- the standard deviation σ is **larger**.

Note the asymmetry you just plotted: s enters **linearly** (double the spread, double the width) but n enters through $\sqrt{n}$ (quadruple the data, halve the width).

What does “95% confidence” mean?

You built one interval from one sample. The “95%” is not about that interval — it is about the **procedure** that produced it.

There is exactly one way to see this: build the interval **many times**, from a population whose true mean you already know, and count.



Exercise 7 — Build 100 intervals and count

Invent a population with a **known** mean: $\mu = 1000$, $\sigma = 850$. Draw 100 samples of $n = 30$ from it, build a 95% interval from each, and count how many of the 100 intervals actually contain 1000. Then plot them.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 set.seed(1)
2 mu_true <- 1000      # the truth -- known only because we invented the population
3 sigma    <- 850
4 n        <- 30
5
6 lo <- numeric(100)
7 hi <- numeric(100)
8
9 for (i in 1:100) {
10   samp <- rnorm(n, mean = mu_true, sd = sigma)
11   # build the 95% interval from THIS sample and store its endpoints in lo[i], hi[i]
12
13 }
14
15 # covers <- ...      TRUE when the interval contains mu_true
16 # sum(covers)        how many out of 100?
17
18 # Picture: one horizontal segment per interval, a vertical line at mu_true.
19 # plot(range(c(lo, hi)), c(1, 100), type = "n", xlab = "price", ylab = "sample")
20 # segments(lo, 1:100, hi, 1:100, col = ifelse(covers, "grey60", "red"), lwd = 2)
```

```
21 # abline(v = mu.true, col = "blue", lwd = 2)
```

95% coverage across repeated samples

- Draw many samples, compute $\bar{x}$ and its interval for each.
- **95% of those intervals cover the true mean.** That is what “95% confidence” means.
- The red intervals in your plot are not mistakes in the arithmetic — they are the 5% the method is *designed* to miss.

This site is archived for reference

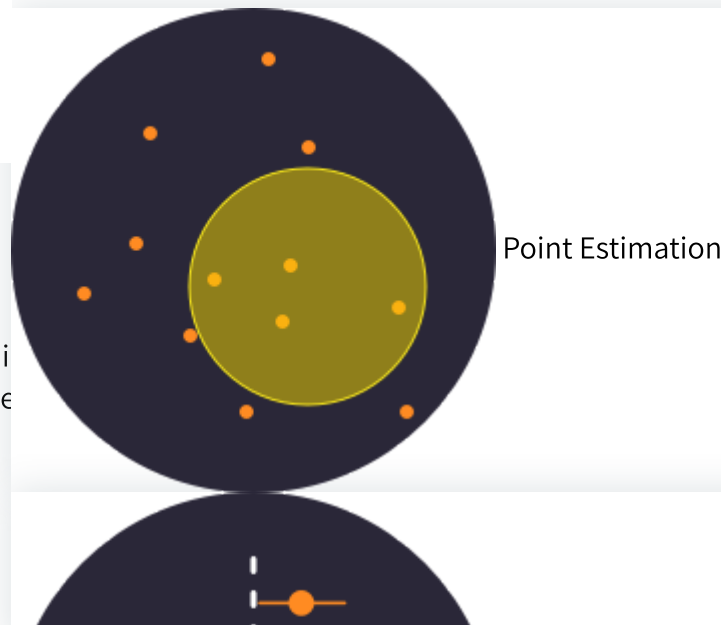
English



Chapter 4: Frequentist Inference

Frequentist Inference

Frequentist inference is the process of determining properties of an underlying distribution via the observation of data.



Source: [Seeing Theory](#)



Exercise 8 — Say it correctly

Your 95% CI for the mean price is {980, 1130}. Which statement is the **correct** interpretation?

- a. We are 95% confident the procedure produced an interval that contains the true mean price.
- b. There is a 95% probability that the true mean price lies between 980 and 1130.
- c. 95% of listing prices fall between 980 and 1130.
- d. If we resampled, the new sample mean would be in this interval 95% of the time.

(a) is correct.

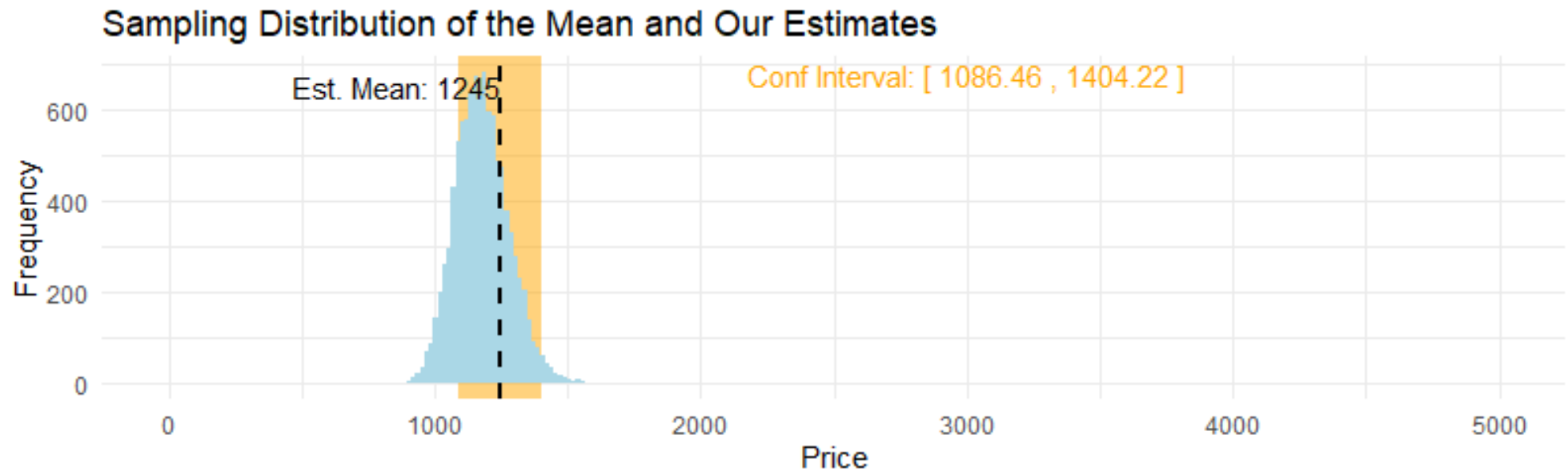
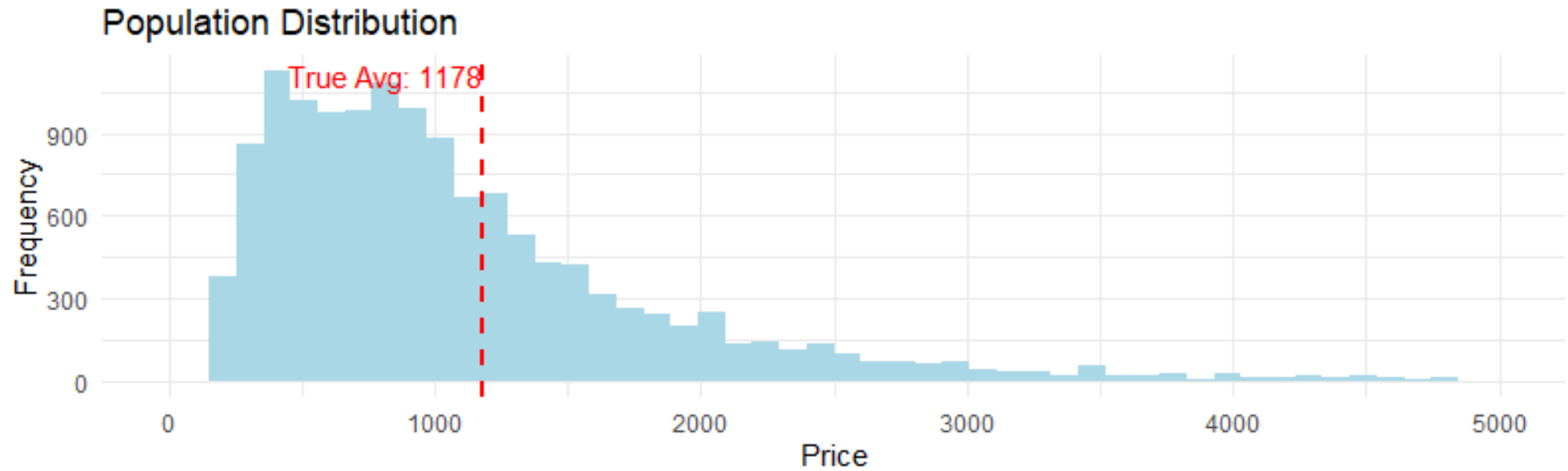
(b) is the one almost everyone picks. In your simulation, once an interval was drawn it was either grey or red — nothing probabilistic left about it. (c) confuses the spread of *prices* with the spread of the *mean*: $sd(x) = 857$ vs $se = 61$. (d) describes a prediction interval for a future $\bar{x}$, not a statement about μ .

Interpreting a confidence interval

$$CI_{90} = \{1086.64, 1404.22\}$$

- **Correct:** we are **90% confident** the interval captures the true mean — i.e. that the true mean price is between 1086.64 and 1404.22.
- **Incorrect:** “with 90% probability the true mean is in this interval.”
 - The computed interval is fixed and the true mean is fixed — no randomness left to attach a probability to.
 - The interval is random only **before** we draw the sample. After computing, the true mean is either in it or not.

The true Airbnb mean



When do we use *normal* critical values?

1. X is not normal:

- $n > 30$ — use the normal (use $n > 40$ if σ unknown, so s approximates σ).
- $n < 30$ — normal approximation is unreliable; you need specialized methods (e.g. bootstrap).

2. X is normal:

- Know $\sigma \rightarrow$ use the normal, any n (with $\sigma, \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$).
- Don't know σ but $n > 40 \rightarrow$ normal (CLT kicks in).
- Don't know σ and $n < 40 \rightarrow$ use **Student's t** (s is a poor estimate of σ at low n).

What's Student's t?

If $X_1, \dots, X_n$ are i.i.d. from $N(\mu, \sigma)$, then

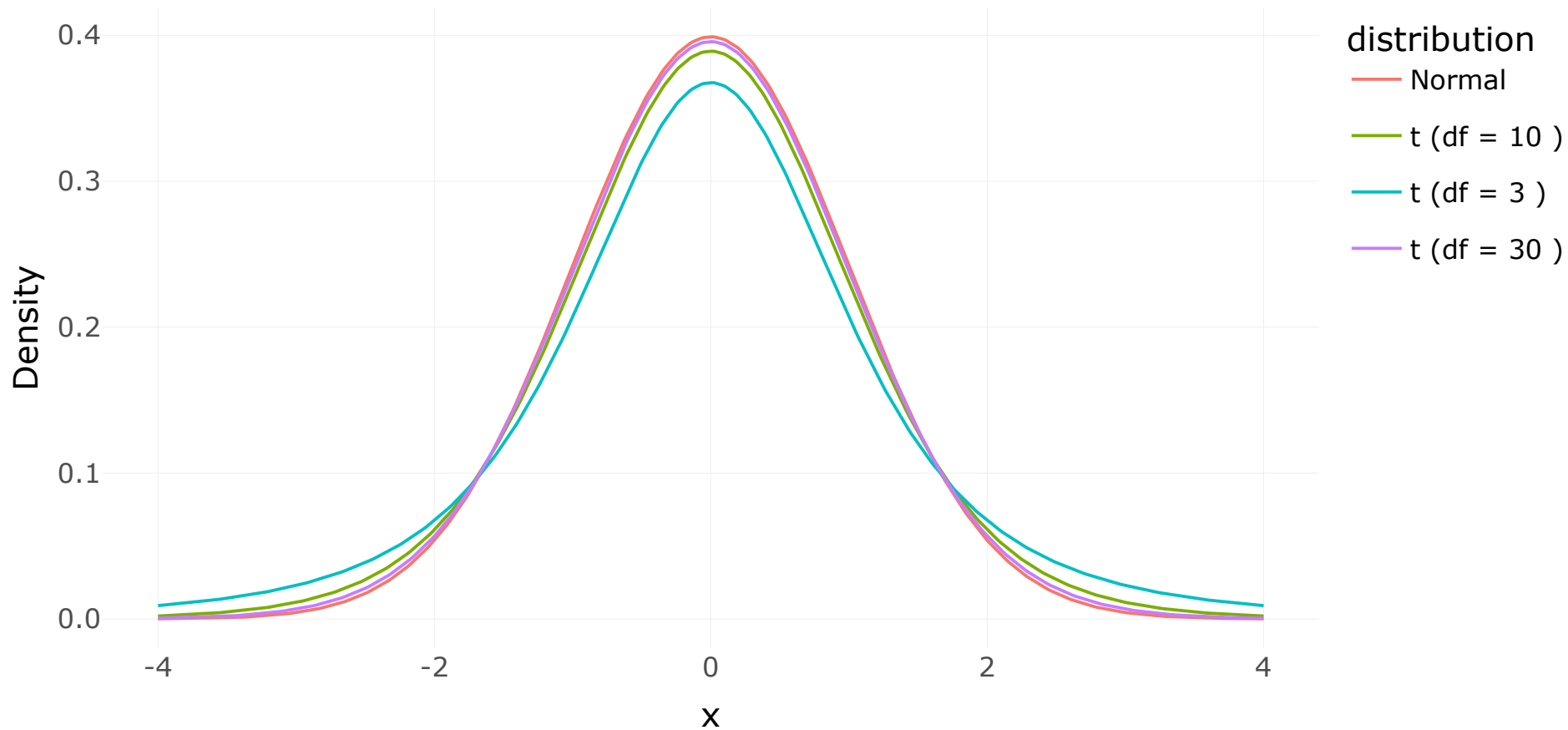
$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

with s the sample standard deviation, follows a **Student's t** distribution with $n - 1$ degrees of freedom:

$$T \sim t_{n-1}$$

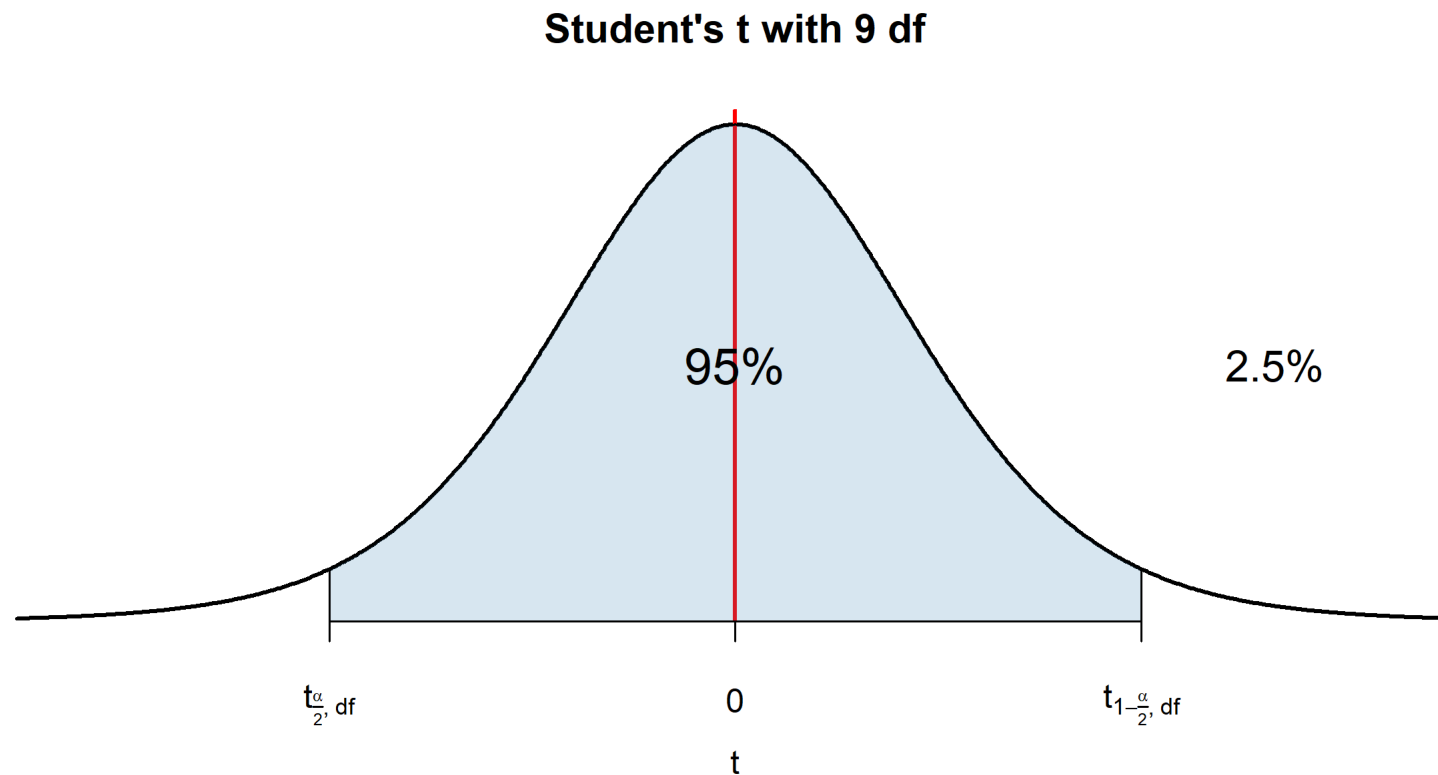
Student's t vs the normal

- Bell-shaped and symmetric around 0.
- **Heavier tails** than the normal — more uncertainty, because we don't know σ .
- Shape set by the degrees of freedom: as n grows, $t_{n-1} \rightarrow N(0, 1)$ (CLT again).



Student's t critical values

1. Get the degrees of freedom: $n - 1$.
2. Get the confidence level $1 - \alpha$, hence $\alpha/2$.
3. Find $t_{1-\frac{\alpha}{2}, n-1}$ with $P(T < t_{1-\frac{\alpha}{2}, n-1}) = 1 - \frac{\alpha}{2}$.



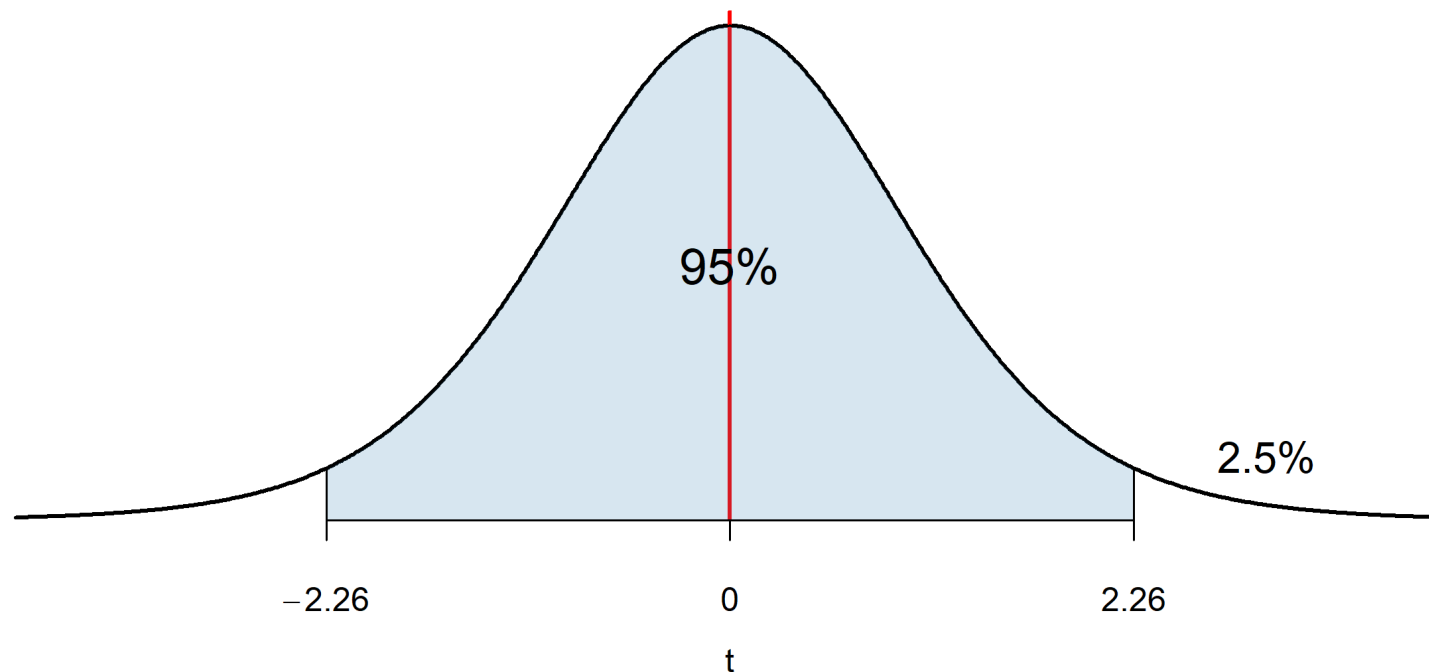
In R: `qt(0.975, 9)` for a 95% CI with 9 df.

t critical value: example

- $n = 10 \rightarrow df = 9$; confidence level 95% $\rightarrow \frac{\alpha}{2} = 0.025$.
- $t_{0.975,9} = 2.26$, then build the CI as before:

$$\left\{ \bar{x} - t_{1-\frac{\alpha}{2},n-1} \cdot \frac{s}{\sqrt{n}}, \bar{x} + t_{1-\frac{\alpha}{2},n-1} \cdot \frac{s}{\sqrt{n}} \right\}$$

Student's t with 9 df



The rule says $n < 40$. Where does that come from?

Using `qnorm` when you should have used `qt` makes your interval too narrow.

At what sample size does the mistake stop mattering? Pick a number before you find out.



Exercise 9 — Find the crossover

Compute both 95% critical values — `qt(0.975, n - 1)` and `qnorm(0.975)` — for a range of sample sizes, and report how far apart they are **in percent**. Find the smallest `n` in your grid where the gap drops below 5%, and below 1%.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 ns <- c(2, 5, 10, 20, 30, 40, 50, 100, 500, 1000)
2
3 # t critical value at each n, the (single) normal critical value,
4 # and the percentage by which t exceeds z.
```

Where the crossover lands

- Below $n \approx 20$ the difference is large and t is not optional.
- By $n = 30$ the t value is ~4% above z ; by $n = 40$, ~3%; by $n = 100$, ~1%.
- That is the whole justification for the $n > 40$ rule of thumb — nothing deeper.

But: using qt is never wrong. It converges to $qnorm$ by itself. When σ is unknown, defaulting to t costs you nothing.



Exercise 10 — A small sample

Take only the first 12 prices, `xs <- x[1:12]`, and build a 95% interval for the mean. Build it **twice** — once with `qnorm`, once with `qt` — and report both widths.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 xs <- x[1:12]
2
3 # Same recipe, on xs. Which critical value belongs here, and what does the other one cost you?
```



Exercise 11 — Did free shipping change spending?

Free shipping was offered to a random group of customers. Spend per customer ($n = 10$):

157.80, 192.45, 210.20, 175.60, 198.30, 180.90, 205.75, 185.20, 177.40, 195.60

- Compute the 90% CI. Which critical value, and what must you assume?
- Average spending **without** free shipping is \$182. Can you conclude free shipping changed spending?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 spend <- c(157.80, 192.45, 210.20, 175.60, 198.30,  
2           180.90, 205.75, 185.20, 177.40, 195.60)  
3  
4 # (a) 90% interval  
5  
6 # (b) is 182 inside it?
```



Exercise 12 — The shortcut, and its default

`t.test()` prints a confidence interval. Get the 95% interval for `x` out of it, then find out what `t.test()` does when you ask for a different confidence level, and check its answer against your Exercise 5 result.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # t.test(x) -- what confidence level does it use by default?
```

```
2
```

```
3 # Get a 99% interval out of it and compare with your own qnorm/qt calculation.
```

Exercise 13 — The whole week, one question

You are the analyst for a host with listings in this market. The city tourism board has published a report stating that

“the average nightly price in this market is 900 pesos.”

Your sample is x . Answer with the tools of this week: **a number, a picture, and one sentence.**

Exercise

 Start Over

 Show Hint

 Run Code

```
1 # 1. A number: the range of values for the true mean that your sample supports.
2 #   Which critical value, and why that one at this n?
3
4 # 2. A picture: something the host can read in three seconds -- where the data sit,
5 #   where your interval sits, where 900 sits.
6
7 # 3. One sentence: what do you tell the host about the board's claim?
```

Exam question — Fall 2025 Midterm 1

Real questions from last year's Midterm 1 — **Households & Environment Survey** ($n = 14,505$ households, each surveyed once, 2017). **We solve these together in class** using `households_environment.rda` (object `exam_data`), in your **Week 5** folder.

Q7. A transportation agency wants to estimate average **weekly household gasoline spending** ($n = 14,505$). Construct a **90% confidence interval** for the mean and report the **lower bound**.

Q8. What happens to the interval if you change the confidence level to **99%**? (no calculation)

Large $n \rightarrow$ use the **normal** critical value. Higher confidence $\rightarrow$ **wider** interval, so a **lower** lower bound.

Practice now (this week's pack): the CI-construction and confidence-level problems in the Week 5 set, then re-run Q7 on `exam_data`.

[→ More exam practice at the end of this deck](#)

What you must be able to do this week

- **Construct a confidence interval for a mean** — compute $\bar{x}$, s , the standard error, and the **margin of error**, then $\bar{x} \pm \text{ME}$.
- Choose the right critical value: **normal** (`qnorm`, large n) vs **Student's t** (`qt`, small n , σ unknown).
- **Interpret** a CI correctly — confidence about the procedure, *not* a probability about a fixed interval.
- Explain what makes an interval **wider or narrower**: confidence level, sample size, and spread.
- Build a **confidence interval for a variance** with the (asymmetric) **chi-square** quantiles (`qchisq`).

Next: from estimating parameters to **testing claims** about them — hypothesis testing.

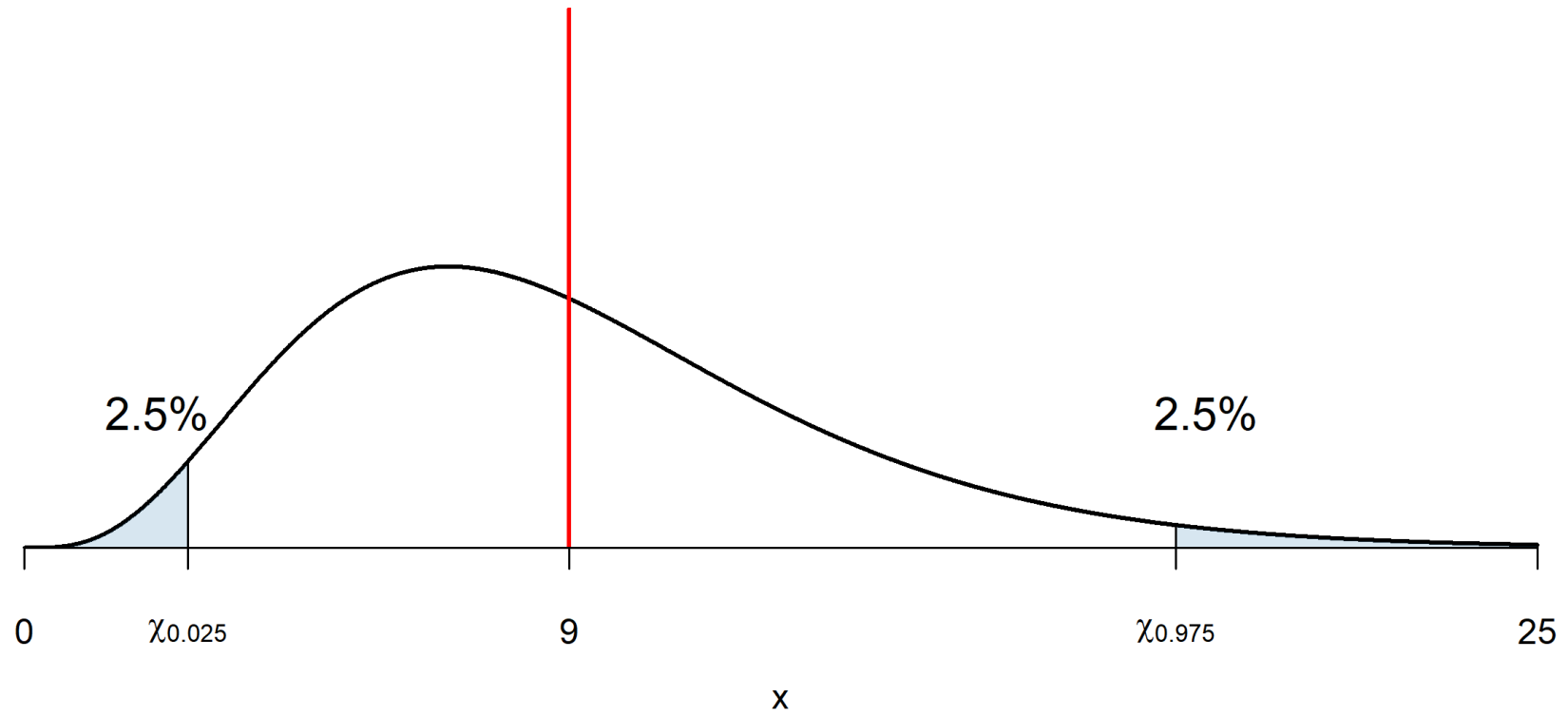
Appendix — optional theory refresher

Confidence intervals for variance

- Sometimes we care about **spread**, not the mean.
- A machine making bolts with high diameter **variance** is unreliable even if the average is correct.
- For $X_1, \dots, X_n$ from a normal, the sample variance $S^2 = \frac{\sum_i (X_i - \bar{X})^2}{n-1}$ satisfies:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2, \quad P\left(\chi_{0.025, n-1}^2 < \frac{(n-1)S^2}{\sigma^2} < \chi_{0.975, n-1}^2\right) = 0.95$$

Chi-square with 9 df



CI for variance: construction

Invert the chi-square statement to isolate σ^2 :

$$\begin{aligned} 0.95 &= P \left(\chi_{0.025, n-1}^2 < \frac{(n-1)S^2}{\sigma^2} < \chi_{0.975, n-1}^2 \right) \\ &= P \left(\frac{(n-1)S^2}{\chi_{0.975, n-1}^2} < \sigma^2 < \frac{(n-1)S^2}{\chi_{0.025, n-1}^2} \right) \end{aligned}$$

So, generally:

$$CI_{1-\alpha} = \left\{ \frac{(n-1)S^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2}, \frac{(n-1)S^2}{\chi_{\frac{\alpha}{2}, n-1}^2} \right\}$$

- The χ^2 quantiles come from R via `qchisq(alpha, df)`. Note the interval is **asymmetric** (chi-square is skewed).



Bonus exercise — Variance CI with `qchisq()`

Quality control: you measure fat content in sausages. From a random sample of **12** sausages you find a sample variance of **20 grams²**. Build the **99% confidence interval for the population variance**, and state what you had to assume.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 n <- 12
2 s2 <- 20
3 # Step 1: the two critical values with qchisq()
4
5 # Step 2: the interval -- which quantile belongs in the LOWER bound?
6
7 # Step 3: is the interval symmetric around s2 = 20? Check the two half-widths.
```

Exam practice

These problems mirror this week's exercise set and the exam — solve each one right here (the **Solution** button reveals the full worked solution).

[← Back to the exam spotlight](#)



Exam practice 1 — Stock returns CI

You are considering investing in a Mexican company listed on the *Bolsa Mexicana de Valores*. Its annual returns over the last 10 years are: 0.03, − 0.12, 0.15, 0.04, − 0.14, − 0.06, 0.00, 0.08, 0.20, 0.07. Assume returns are i.i.d. normal. [13]

- What is the 95% CI for μ ?
- What is the t -statistic for $H_0 : \mu = 0$? Do you reject at 5%?
- What is the p-value?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 returns <- c(0.03, -0.12, 0.15, 0.04, -0.14, -0.06, 0.00, 0.08, 0.20, 0.07)
2
3 # (a) the interval
4
5 # (b) the t-statistic, and the critical value you compare it to
6
7 # (c) the p-value
```

- How would the t -statistic for $H_0 : \mu = -0.01$ compare to (b)?

Larger in absolute value: $t = \frac{0.025 - (-0.01)}{0.0346} = 1.012$, because -0.01 is farther from $\bar{x} = 0.025$ than 0 is.

- How would the p-value compare to (c)?

Exam practice 2 — Government bond returns

Monthly returns on a Mexican government bond are i.i.d. from a normal distribution with unknown mean and standard deviation. You observe **12 months**, with sample average **0.012** and standard deviation **0.07**. [111]

- Provide 95% and 99% confidence intervals for μ_X using the t distribution.
- Compare these intervals to those based on $N(0, 1)$.
- Suppose instead $n = 100$, same sample average and standard deviation. Provide 95% and 99% CIs using the normal approximation.

Exercise

 Start Over

 Show Hint

 Run Code

```
1 xbar <- 0.012; s <- 0.07; n <- 12
2
3 # (a) two t-based intervals
4
5 # (b) the same two with normal critical values -- which are wider, and by how much?
6
7 # (c) n = 100, normal approximation
```



Exam practice 3 — Employee productivity

A random sample of 25 employees at a firm in Mexico City yields $\bar{x} = 72$ (standardized productivity test) and $s_x = 15$. Assume test scores are normally distributed with unknown mean and variance. [112]

- Provide 90% and 95% confidence intervals for μ_X .
- Compare these to intervals based on $N(0, 1)$.

Exercise ↺ Start Over 💡 Show Hint ▶ Run Code

```
1 xbar <- 72; s <- 15; n <- 25
2
3 # (a) 90% and 95%, t-based
4
5 # (b) the same two, normal-based
```

- If instead $\bar{x} = 76$, $s_x = 15$, how would the width of the 95% CI compare?

Unchanged — width = $2 \times 2.064 \times 3 = 12.38$. Changing $\bar{x}$ shifts the interval's *center* but not its width; width depends only on s , n , and the critical value.

- If instead $\bar{x} = 72$, $s_x = 20$, how would the width of the 95% CI compare?

Wider — $SE = 20/5 = 4$, width = $2 \times 2.064 \times 4 = 16.51$. Greater variability means more uncertainty about the population mean.



Exam practice 4 — Reverse-engineer the CI

A Mexican company's weekly profits are i.i.d. draws from $X \sim N(\mu, \sigma^2)$. After n weeks, two confidence intervals for μ are constructed — **95% CI:** (9.201, 14.799) and **90% CI:** (9.955, 14.045). [I13]

- What is the sample average of weekly profits?
- What is n ? (assume the Student's t distribution was used)
- What is the sample standard deviation of weekly profits?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # (a) what feature of an interval is the sample average?
2
3 # (b) the two intervals share a center and a SE -- only the critical value differs.
4 #     Search over df: which one reproduces the ratio of half-widths?
5
6 # (c) back out s from the half-width
```



Exam practice 5 — Rescale a confidence interval

The two-sided asymptotic **95%** confidence interval for a parameter is $(12, 22)$. ^[15]

a. What is the two-sided asymptotic **80%** confidence interval for the same parameter?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # The interval is all you get. Recover its two ingredients first, then rebuild.
```



Exam practice 6 — Defective computers (proportion)

A university IT department in Mexico has many old computers, some no longer working. To estimate the proportion defective, staff test **80** randomly and find **20** defective. Find the asymptotic 95% CI for the population proportion. [16]

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 n <- 80; x <- 20
2
3 # A proportion is a mean of a 0/1 variable -- but its SE has its own formula.
```



Exam practice 7 — Gardening promotion

A gardening company offers a promotion: n households get a free first-time service, after which some continue as paying customers. The probability of continuing is π . After observation, the asymptotic 95% confidence interval for π is $(0.1800, 0.3200)$. ^[17]

- a. What is the sample proportion?
- b. What is n ?

Exercise

 Start Over

 Show Hint

 Run Code

```
1 # (a) read the sample proportion off the interval
2
3 # (b) the half-width contains n -- invert the proportion-SE formula to get it out
```



Exam practice 8 — Sample size for a target width

A marketing firm wants to estimate the average daily hours of streaming (Netflix/Disney+) per household in Mexico. Suppose $\sigma = 2.5$ hours. They want the **99%** confidence interval **width** to be no larger than **0.6** hours. What is the minimum required sample size? [118]

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 sigma <- 2.5
2
3 # Write the width as a function of n, set it equal to the target, solve for n.
4 # Watch two things: width vs half-width, and rounding in the right direction.
```



Exam practice 9 — Union membership by gender

The `cps` dataset (package `probstats4econ`) contains U.S. labor-force data with `gender` and `unionstatus`. The two-way table `table(cps$gender, cps$unionstatus)` is loaded here as `cps_tab`. [119]

- Get the union **rate** within each gender out of the table.
- Compute the asymptotic 95% confidence interval for the probability that a **male worker** is in a union.
- Compute the asymptotic 95% confidence interval for the probability that a **female worker** is in a union.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 cps_tab
2
3 # (a) the raw counts are not comparable -- there are more men. Turn each ROW into shares.
4
5 # (b), (c) each rate is a proportion with its own n: the ROW total.
```



Exam practice 10 — Strike durations

The `strikes` dataset (package `probstats4econ`; 566 strike durations in weeks, 1968–1976) has key variable `duration`. Part (a) asks for the mean, median, and standard deviation — at home run `mean(strikes$duration)` etc.; the results are: **mean = 43.624, median = 28, sd = 44.666.** [120]

b. Construct a 90% confidence interval for the average strike duration.

Exercise ↺ Start Over 💡 Show Hint ▶ Run Code

```
1 xbar <- 43.624; s <- 44.666; n <- 566
2
3 # The 90% interval. Which critical value at this n?
4
5 # Then: the interval is a few weeks wide, but sd is 44.7 weeks.
6 # Compare the two numbers -- what is each one describing?
```

The mean (43.6) is well above the median (28). What does that tell you about the distribution?

Right-skewed — a few very long strikes pull the mean up above the median.